Section 1 – MCQs Answers

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1. Which OSI layer is responsible for logical addressing?

✅ **Correct Answer: B) Network**

Explanation:
The **Network Layer (Layer 3)** is responsible for **logical addressing**, which means **IP addressing**.
Routers work at this layer and use IP addresses to send packets between networks.

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2. Which protocol is connection-oriented?

✅ **Correct Answer: C) TCP**

Explanation:
TCP (Transmission Control Protocol) is **connection-oriented**, meaning it establishes a connection before sending data.
It ensures:

- Reliable communication
- Data delivery
- Correct order of packets

UDP is **connectionless**.

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3. What is the default administrative distance of OSPF?

✅ **Correct Answer: C) 110**

Explanation:

Protocol	Administrative Distance
Connected	0
Static	1

Protocol	Administrative Distance
EIGRP	90
OSPF	**110**
RIP	120

Administrative Distance (AD) decides **which route is more trusted**.

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4. Which command is used to configure a standard ACL in Cisco?

✅ **Correct Answer: C) access-list 10 permit**

Explanation:
Standard ACL numbers range from **1–99**.

Example:

access-list 10 permit 192.168.1.0 0.0.0.255

Standard ACL filters **only source IP address**.

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5. Which protocol is used to resolve IP to MAC address?

✅ **Correct Answer: B) ARP**

Explanation:
ARP (Address Resolution Protocol) converts **IP address → MAC address**.

Example process:

Computer wants to send packet to **192.168.1.10**
First it finds **MAC address** using ARP.

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6. What is the purpose of FHRP?

✅ **Correct Answer: B) Provide gateway redundancy**

Explanation:
FHRP (First Hop Redundancy Protocol) provides **backup gateway** if the main gateway fails.

Examples:

- HSRP
- VRRP
- GLBP

This ensures **high availability** in network.

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7. HSRP is developed by which company?

✅ **Correct Answer: C) Cisco**

Explanation:
HSRP (Hot Standby Router Protocol) is a **Cisco proprietary protocol** used for **gateway redundancy**.

One router is **active**, another is **standby**.

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8. Which routing protocol is link-state?

✅ **Correct Answer: B) OSPF**

Explanation:

Protocol	Type
RIP	Distance Vector
IGRP	Distance Vector
OSPF	**Link-State**
BGP	Path Vector

OSPF uses **Shortest Path First algorithm (Dijkstra)**.

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9. Extended ACLs filter traffic based on:

✅ **Correct Answer: C) Source and destination IP and ports**

Explanation:
Extended ACL can filter using:

- Source IP
- Destination IP
- Protocol (TCP/UDP/ICMP)
- Port numbers

Example:

access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 80

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10. Which multicast address is used by OSPF?

✅ **Correct Answer: A) 224.0.0.5**

Explanation:

OSPF uses two multicast addresses:

Address	Purpose
224.0.0.5	All OSPF routers
224.0.0.6	DR/BDR routers

Routers use this for exchanging **LSA updates**.

Section 2 – Short Answer Questions

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11. What is the difference between TCP and UDP?

TCP and UDP both are **Transport Layer protocols** but they work in different way.

TCP (Transmission Control Protocol)

- It is **connection-oriented** protocol.
- It makes connection before sending data.
- Data is sent in **proper sequence**.
- It provides **reliable communication**.
- If packet lost, TCP **retransmits the packet**.

Example uses:
Web browsing, Email, File transfer.

UDP (User Datagram Protocol)

- It is **connectionless protocol**.
- It does **not establish connection** before sending data.
- Data may not arrive in sequence.
- It is **faster but less reliable**.

Example uses:
Video streaming, VoIP, Online gaming.

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12. What is the function of a default gateway?

Default gateway is the **router IP address used to send traffic outside the local network**.

When a device wants to communicate with another device **in different network**, it sends the packet to the **default gateway**.

Example:

PC IP
192.168.1.10

Default Gateway
192.168.1.1

If destination network is not local, packet goes to **router (default gateway)**.

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13. What is administrative distance?

Administrative Distance (AD) is a **value used by router to decide which route is more trustworthy** when there are multiple routes to the same network.

Lower AD means **more preferred route**.

Example:

Route Type	AD
Connected	0
Static	1
EIGRP	90
OSPF	110
RIP	120

Router always selects **route with lowest AD**.

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14. What is the difference between static and dynamic routing?

Static Routing

- Routes are **manually configured by network administrator**.
- Suitable for **small networks**.
- Less CPU usage.
- But difficult to manage in large networks.

Example command:

ip route 192.168.2.0 255.255.255.0 10.0.0.2

Dynamic Routing

- Router **automatically learns routes** using routing protocols.
- Used in **large networks**.
- Automatically updates when network changes.

Examples of protocols:

- RIP
- OSPF
- EIGRP
- BGP

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15. What is a wildcard mask in ACL?

Wildcard mask is used in **Access Control List (ACL)** to define **which IP addresses should match**.

It is **opposite of subnet mask**.

Rule:

- **0 → must match**
- **1 → ignore**

Example:

access-list 10 permit 192.168.1.0 0.0.0.255

Meaning:

Allow all IPs from
192.168.1.1 – 192.168.1.254

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16. Where should you place a standard ACL and why?

Standard ACL should be placed **close to the destination network**.

Reason:
Standard ACL filters **only source IP address**, it does not check destination IP.

If we place it near source, it may block traffic that should go to other networks.

So best practice is **place standard ACL near destination**.

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17. What is the difference between HSRP and VRRP?

Both protocols provide **default gateway redundancy**.

HSRP (Hot Standby Router Protocol)

- Developed by **Cisco**
- Cisco proprietary protocol
- One router active, another standby
- Virtual IP used as gateway

VRRP (Virtual Router Redundancy Protocol)

- **Open standard protocol**
- Works with different vendors
- Similar function as HSRP
- One router is master, others backup

Main difference:
HSRP = Cisco proprietary
VRRP = Open standard

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18. What is the purpose of hello packets in OSPF?

Hello packets are used to:

1. **Discover neighbor routers**
2. **Maintain neighbor relationship**
3. **Check if neighbor router is alive**
4. **Elect DR and BDR in broadcast networks**

If hello packets stop coming, router assumes **neighbor is down**.

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19. What is passive-interface command used for?

passive-interface command is used to **stop sending routing updates on a specific interface**.

Router will:

- **Not send routing updates**
- But still **advertise that network**

Example:

router ospf 1
passive-interface g0/1

Used when:

- Interface connected to **end devices**
- To improve **network security**
- Reduce unnecessary routing traffic

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20. What is the difference between standard ACL and extended ACL?

Feature	Standard ACL	Extended ACL
Filter based on	Source IP only	Source IP + Destination IP
Protocol filtering	No	Yes

Feature	Standard ACL	Extended ACL
Port filtering	No	Yes
ACL number range	1 – 99	100 – 199
Placement	Near destination	Near source

Example:

Standard ACL

access-list 10 permit 192.168.1.0 0.0.0.255

Extended ACL

access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 80

Section 3 – Definitions

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21. Define IP Address

An **IP Address (Internet Protocol Address)** is a **unique logical address assigned to each device in a network** so that devices can **identify and communicate with each other**.

It works like a **home address for a device on the network**.

Example:

- IPv4 → 192.168.1.10
- IPv6 → 2001:db8::1

Routers use IP addresses to **route packets from source to destination**.

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22. Define Subnetting

Subnetting is the process of **dividing a large network into smaller logical networks called subnets**.

Purpose of subnetting:

- Better **network management**
- Reduce **broadcast traffic**
- Improve **network performance**
- Increase **security**

Example:

Network:
192.168.1.0/24

Can be divided into multiple smaller networks like:

192.168.1.0/26
192.168.1.64/26
192.168.1.128/26

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23. Define Routing Table

A **Routing Table** is a table stored in a **router or network device** that contains information about **network paths and destinations**.

Router checks the routing table to decide **where to forward the packet**.

Routing table contains:

- Destination network
- Next hop
- Interface
- Metric

Example command to view:

show ip route

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24. Define ACL (Access Control List)

ACL (Access Control List) is a set of rules used in routers or switches to **permit or deny network traffic**.

ACL is mainly used for:

- Network **security**
- **Traffic filtering**
- **Access control**

Two main types:

- **Standard ACL** – filters using source IP
- **Extended ACL** – filters using source IP, destination IP, protocol and ports

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25. Define FHRP (First Hop Redundancy Protocol)

FHRP (First Hop Redundancy Protocol) is used to provide **backup gateway in a network**.

If the **main gateway router fails**, another router automatically takes over so that **network communication continues without interruption**.

Examples of FHRP:

- HSRP
- VRRP
- GLBP

Purpose: **High availability and gateway redundancy**

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26. Define HSRP

HSRP (Hot Standby Router Protocol) is a **Cisco proprietary protocol** used to provide **default gateway redundancy**.

In HSRP:

- One router acts as **Active Router**
- Another router acts as **Standby Router**

Both routers share a **virtual IP address** which is used by hosts as the **default gateway**.

If the active router fails, the standby router becomes active.

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27. Define VRRP

VRRP (Virtual Router Redundancy Protocol) is a **network protocol that provides gateway redundancy**, similar to HSRP.

Difference:

- VRRP is an **open standard protocol**
- Works with **different vendor devices**

In VRRP:

- One router is **Master**
- Others are **Backup**

If the master router fails, backup router takes over automatically.

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28. Define OSPF

OSPF (Open Shortest Path First) is a **link-state dynamic routing protocol** used in large networks.

It calculates the **shortest path to destination using Dijkstra algorithm**.

Key features:

- Fast convergence
- Uses **cost metric**
- Supports **hierarchical design with areas**
- Uses **link-state advertisements (LSA)**

Default Administrative Distance of OSPF = **110**

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29. Define Convergence

Convergence in networking means the **time taken by routers to update their routing tables and reach a consistent network topology after a change in the network**.

Example of network change:

- Link failure
- Router failure
- New route added

After convergence, all routers have **updated and accurate routing information**.

Faster convergence means **better network performance and stability**.

Section 4 – Practical

30. OSPF Configuration Practical

Network Topology

Three routers connected in a line:

R1 —— R2 —— R3

Networks:

R1 → 192.168.1.0/24
R2 → 192.168.2.0/24
R3 → 192.168.3.0/24

All routers will run **OSPF Process ID 1** and advertise networks in **Area 0**.

Step 1 – Configure IP Address on Router Interfaces

First assign IP addresses on router interfaces.

R1 Configuration

```
enable
configure terminal

interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown

interface g0/1
ip address 10.0.12.1 255.255.255.252
no shutdown
```

R2 Configuration

```
enable
configure terminal

interface g0/0
ip address 10.0.12.2 255.255.255.252
```

```
no shutdown

interface g0/1
ip address 10.0.23.1 255.255.255.252
no shutdown

interface g0/2
ip address 192.168.2.1 255.255.255.0
no shutdown
```

R3 Configuration

```
enable
configure terminal

interface g0/0
ip address 10.0.23.2 255.255.255.252
no shutdown

interface g0/1
ip address 192.168.3.1 255.255.255.0
no shutdown
```

Step 2 – Configure OSPF Process ID 1

Now enable OSPF on all routers.

R1 OSPF Configuration

```
router ospf 1

network 192.168.1.0 0.0.0.255 area 0
network 10.0.12.0 0.0.0.3 area 0
```

Explanation:

- router ospf 1 → start OSPF process

- network command → advertise networks in OSPF
- area 0 → backbone area

R2 OSPF Configuration

router ospf 1

network 192.168.2.0 0.0.0.255 area 0
network 10.0.12.0 0.0.0.3 area 0
network 10.0.23.0 0.0.0.3 area 0

R3 OSPF Configuration

router ospf 1

network 192.168.3.0 0.0.0.255 area 0
network 10.0.23.0 0.0.0.3 area 0

Step 3 – Verify OSPF Neighbors

After configuration we check if routers formed neighbor relationship.

Command:

show ip ospf neighbor

Expected result:

R1 should see **R2 as neighbor**
R2 should see **R1 and R3**
R3 should see **R2**

State should show **FULL**.

Step 4 – Verify Routing Table

Now check if routes are learned through OSPF.

Command:

show ip route

In routing table you should see **O** symbol.

Example:

O 192.168.2.0/24
O 192.168.3.0/24

Meaning:

Routes learned through **OSPF protocol**.

Step 5 – Test Connectivity

Use ping command to test connectivity between networks.

Example:

ping 192.168.3.1

If ping is successful → OSPF routing is working correctly.

Final Result

After configuration:

- All routers run **OSPF Process ID 1**
- All networks are in **Area 0**
- Routers form **neighbor relationship**
- Routing table contains **OSPF learned routes**
- All networks can communicate